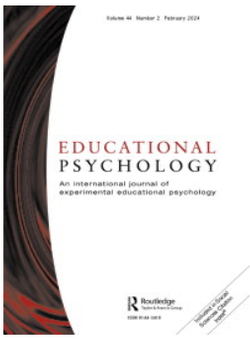

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Social-emotional need satisfaction and students' academic engagement and social-emotional skills

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ABSTRACT

This investigation examined the role of students' social-emotional need satisfaction in relation to academic engagement and social-emotional skills. Among a sample of 501 secondary school students (and their parents/carers), several need satisfaction variables were examined: perceived social-emotional autonomy, perceived social competence, perceived emotional competence, perceived relatedness with students, and perceived relatedness with teachers. The hypothesised outcomes were student-reported behavioural disaffection, and parent reports of students' homework practices, expressive skill, and perspective-taking skill. Results demonstrated that perceived social competence was associated with lower behavioural disaffection. Perceived emotional competence was associated with more positive homework practices, greater expressive skill, and greater perspective-taking skill. Perceived relatedness with teachers was associated with lower behavioural disaffection and more positive homework practices. Findings have implications for supporting students' positive adjustment in school and beyond.

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Social-emotional competence; basic psychological need satisfaction; academic engagement; social-emotional skills

Introduction

A vast body of research highlights the salience of basic psychological need satisfaction for positive individual outcomes (Ryan & Deci, 2017). Most of the research on need satisfaction among students has involved examining need satisfaction related to the school context (Jang et al., 2016; Theis et al., 2020). Recently, emerging work has considered the role of social-emotional need satisfaction, which reflects students' perceived social-emotional autonomy (i.e. having choice and control over how they think, act, and feel), competence (i.e. feeling able to effectively navigate social and emotional interactions), and relatedness (i.e. feeling connected to and accepted by others; Collie, 2020). This research has shown that social-emotional need satisfaction is linked with

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positive social-emotional outcomes (e.g. autonomous prosocial motivation, well-being; Collie, 2022c). However, important gaps in knowledge remain. Although there is ample research looking at relatedness with students and teachers (e.g. Engels et al., 2021; Guo et al., 2023; Santos et al., 2023), and growing research examining perceived social-emotional competence (Caprara et al., 2008; Kristensen et al., 2021), limited research has considered perceived social-emotional autonomy and all social-emotional need satisfaction factors simultaneously. Examining the social-emotional needs concurrently is necessary to ascertain the unique role of each factor – and thus to guide what factors should form a focus in efforts to boost different outcomes. Furthermore, previous studies on perceived social-emotional autonomy and competence appear to have focused on links with social-emotional behaviour and well-being. Efforts that extend this understanding to academic engagement and social-emotional skills are now important.

The aim of the present study, therefore, is to examine social-emotional need satisfaction in relation to two academic engagement factors (student-reported behavioural disaffection, and parent reports of students' homework practices) and two social-emotional skills (parent reports of students' expressive skill and perspective-taking skill). Social-emotional need satisfaction was assessed by way of perceived social-emotional autonomy, perceived social competence, perceived emotional competence, perceived relatedness with students, and perceived relatedness with teachers. Figure 1 displays the hypothesised model. Findings have the potential to yield understanding about the role of social-emotional need satisfaction in relation to factors previously not considered, and also to isolate the specific need satisfaction factors that are uniquely relevant to the different academic engagement and social-emotional skill factors.

Conceptual framing

The study was guided by the Social and Emotional Competence (SEC) School Model (Collie, 2020), which is informed by motivational theory (self-determination theory; Ryan & Deci, 2017) and the SEC literature (Rose-Krasnor & Denham, 2009). Most research to date on SEC has focused on social-emotional manifestations (i.e. the outcomes of SEC) by measuring skills and behaviours (Stump et al., 2009). The SEC School Model differs in that it emphasises the mechanisms and manifestations implicated in the development of SEC. Specifically, the model describes how social-emotional outcomes – that is, manifestations of SEC – are promoted by underlying mechanisms

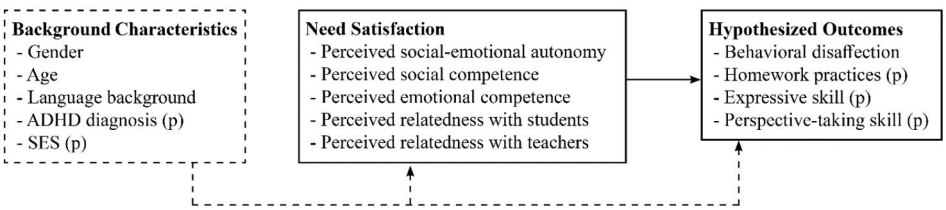

Figure 1. Hypothesised model. Parent-reported factors are represented by (p). All other substantive factors were reported by students. ADHD: attention-deficit/hyperactivity disorder; SES: socio-economic status.

inherent to SEC and its development. One central mechanism in the SEC School Model is social-emotional basic psychological need satisfaction. The fundamental role of need satisfaction in optimal human functioning has been emphasised by motivation theories for decades (Ryan & Deci, 2017). As described in detail below, individuals who experience need satisfaction within a context tend to internally endorse the values and norms of that context, leading to beneficial outcomes such as motivation, engagement, and performance (Ryan & Deci, 2017). This process has been supported by longitudinal and experimental research in the academic domain (e.g. Cheon et al., 2018). The SEC School Model adopts this viewpoint and extends it to the social and emotional domains. More precisely, a focal process in the SEC School Model is that social-emotional need satisfaction is associated with positive social-emotional outcomes, including adaptive behaviours and well-being. The present study involved examining this focal process and extending it by investigating academic outcomes alongside previously unexamined social-emotional skills. The present study, thus, has the potential to yield understanding about the unique role of the social-emotional need satisfaction factors in relation to novel outcomes. Each factor examined in the current study is introduced in the following sections.

Social-emotional need satisfaction

Researchers have established the salience of three basic psychological needs for supporting positive individual outcomes. Perceived autonomy reflects a feeling of being the originator of one's actions (de Charms, 1968). Perceived competence involves a feeling of being effective in one's pursuits (White, 1959). Perceived relatedness involves a feeling of being connected to and cared for by significant others (Baumeister & Leary, 1995). These three psychological needs are integral components of self-determination theory (Ryan & Deci, 2017), which establishes that satisfaction of these needs is linked to positive outcomes. This claim is supported by abundant research in the academic domain, linking these needs to greater academic motivation (Ahn et al., 2021), engagement (Jang et al., 2016), achievement (Theis et al., 2020), and prosocial behaviours (Cheon et al., 2018). In the social-emotional domain, satisfying these needs is also vital (Ryan & Deci, 2017). Whereas there is ample research on relatedness with students and teachers (for reviews, see Tao et al., 2022; Wentzel et al., 2021), research on perceived social-emotional autonomy and competence is limited, as is research examining all social-emotional need satisfaction factors concurrently. The operationalisation of social-emotional need satisfaction according to the SEC School Model is now introduced.

Perceived social-emotional autonomy reflects the perception of having a say in how one thinks, acts, and feels in relation to social and emotional interactions (Collie, 2020; cf. de Charms, 1968). Perceived social-emotional competence broadly denotes an individual's subjective perception of their ability to effectively navigate social and emotional interactions (Collie, 2020; cf. White, 1959). In the present study, perceived social competence and perceived emotional competence were examined as separate factors to yield understanding about their distinct roles. Drawing on Collie's (2022a) work, *perceived social competence* refers to the sense of being able to be assertive (i.e.

to advocate for oneself or others), act with tolerance towards others, and self-regulate one's behaviours to match the requirements in different contexts. *Perceived emotional competence* reflects the sense of being able to adjust one's thinking to feel better (e.g. lower frustration, more happiness), and to be able to recognise the emotions one experiences (Collie, 2022a). Finally, *perceived relatedness with students* and *perceived relatedness with teachers* are inherently social in nature and refer to a sense of connection with, being cared for, acceptance, and respect between the individual and other students or teachers, respectively (Baumeister & Leary, 1995; Ryan & Deci, 2017).

Academic engagement and social-emotional skill

Need satisfaction lays a foundation for positive individual outcomes (Ryan & Deci, 2017). Hypothesised outcomes relevant to behavioural academic engagement and social-emotional skills were examined in the present study. *Behavioural disaffection,* also known as behavioural disengagement, refers to a set of maladaptive behaviours characterised by passivity and withdrawal from learning tasks (Skinner et al., 2008). *Homework practices* refer to the manner in which students perform and complete their homework, encompassing both the quality and quantity of their efforts (Zimmerman & Kitsantas, 2005). *Expressive skill* refers to students' ability to effectively communicate their thoughts and feelings to others (Soto et al., 2022), and *perspective-taking skill* refers to students' ability to understand the thoughts and feelings of others (Soto et al., 2022). Taken together, the hypothesised outcomes are salient for positive academic, social, and emotional adjustment. The two academic outcomes were selected because they capture assessments of students' behavioural (dis)engagement within and outside of the classroom (Gregory & Korth, 2016): behavioural disaffection reflects academic disengagement in the classroom, whereas homework practices reflect academic engagement beyond the classroom. The two social-emotional skills were selected because they capture the ability to understand oneself and others, and are known to be critical for positive adolescent development (Soto et al., 2022). In the current study, student reports were used for behavioural disaffection because this reflects students' reactions (e.g. pretending to work, not trying hard) that are driven by intrapsychic motives. Parent reports were utilised for the remaining outcomes because students' social-emotional need satisfaction likely plays a role in how they would self-report these outcomes. By using parent reports, it was possible to disentangle that and also reduce concerns about common-method bias.

Prior research demonstrates that social-emotional need satisfaction is linked with positive outcomes among students. For example, perceived social-emotional autonomy is associated with lower negative affect (Collie, 2022c). Perceived social competence is linked with lower psychological distress (Kristensen et al., 2021), greater positive affect (Collie, 2022c), and greater self-reported perspective-taking skill (Wentzel et al., 2007). Perceived emotional competence is associated with greater prosocial behaviour, lower conduct problems (Bigman et al., 2016; Caprara et al., 2008; Parise et al., 2019), and greater well-being (Metz et al., 2013). Ample research has shown that perceived relatedness with student and teachers is positively associated

with academic outcomes (including engagement, e.g. Engels et al., 2021; Santos et al., 2023), as well as social-emotional skills (e.g. Guo et al., 2023).

Despite this body of work, researchers have apparently not considered academic outcomes in relation to perceived social-emotional autonomy and competence. Moreover, there is a need to expand knowledge of all need satisfaction factors in relation to social-emotional skills. The aim of the present study was to address these gaps by assessing four novel hypothesised outcomes. Based on the SEC School Model (Collie, 2020), it was anticipated that the factors of social-emotional need satisfaction would be negatively associated with behavioural disaffection, and positively associated with the remaining outcomes. More precisely, perceived social-emotional autonomy at school involves feeling a sense of self-determination (Collie, 2020), which is important for internalising social-emotional and academic norms and values within that context and thus outcomes associated with that (Ryan & Deci, 2017) – such as higher engagement (and lower disengagement) and more adaptive social-emotional outcomes. Perceived social competence and perceived emotional competence mean that students feel capable in their social and emotional interactions, which helps students to mobilise thoughts and actions related to those perceptions (Ryan & Moller, 2017) – likely leading to adaptive engagement and social-emotional skills as a result (Collie, 2022a). Although not directly applicable, this hypothesised association is tentatively supported by research linking social-emotional skills with greater academic engagement and lower disengagement (Hasty et al., 2023; Santos et al., 2023). Finally, perceived relatedness with students and teachers mean that students feel cared for and accepted in the classroom, which is important for academic and social-emotional development (Engels et al., 2021; Guo et al., 2023; Santos et al., 2023).

Relevant background characteristics to consider

Several background characteristics were examined as covariates in the present study: gender, age, language background, attention-deficit/hyperactivity disorder (ADHD) diagnosis, and socioeconomic status (SES). These five characteristics were investigated because prior research shows they are associated with the academic and social-emotional factors examined in the study. Moreover, ADHD is the most prevalent neurodevelopmental disorder among Australian adolescents (e.g. 7.4% prevalence vs. 0.7% prevalence for autism spectrum disorder; Australian Institute of Health and Welfare, 2017; Lawrence et al., 2015). Female students and those from non-English speaking backgrounds have been shown to report greater perceived social-emotional competence (Santos et al., 2023). Older students have been found to report lower social-emotional skills (Guo et al., 2023; Santos et al., 2023). Students with ADHD have been shown to report lower engagement (Vile Junod et al., 2006), and students from higher SES backgrounds have been demonstrated to report more positive homework practices (Martin et al., 2022). Given this, adjusting for these background characteristics in analyses is important for ascertaining unique variance explained by substantive factors.

Study overview

This study involved examining students' social-emotional need satisfaction in relation to academic engagement and social-emotional skills. The following social-emotional need satisfaction factors were examined: perceived social-emotional autonomy, perceived social competence, perceived emotional competence, perceived relatedness with students, and perceived relatedness with teachers. The hypothesised outcomes were student-reported behavioural disaffection, and parent reports of students' homework practices, expressive skill, and perspective-taking skill. Structural equation modelling was used to ascertain the extent to which the need satisfaction factors are linked with the outcomes (controlling for background characteristics). Figure 1 displays the hypothesised model.

Method

Sample and procedure

This study consisted of 501 students (and their 501 parents/carers) from secondary schools across all states/territories in Australia. The students were in grades 7 (14%), 8 (26%), 9 (25%), 10 (21%), 11 (13%), and 12 (1%). The sample was 51% female, 48% male, and 1% who identified as non-binary or other. The average age of the students was 14 ($SD = 1$; range 13–16) years, and the majority spoke only English at home (91%). Additionally, 10.8% of the students had received an ADHD diagnosis. More precisely, 5.2% had predominantly inattentive presentation, 2.0% had predominantly hyperactive-impulsive presentation, and 3.6% had combined presentation. These prevalence statistics are slightly above, but similar in breakdown to national reports (e.g. 7.4% ADHD prevalence rate; Lawrence et al., 2015). In terms of socio-economic status (SES), the average was 1002 ($SD = 68$), which is similar to the national average ($M = 1000$, $SD = 100$; Australian Bureau of Statistics [ABS], 2018). The sample included students from government (66%), Catholic (20%), and independent (14%) schools, with 92% of the schools being co-educational, 4% single-sex girls, and 4% single-sex boys. Regarding parental involvement, female (63%), male (36%), and non-binary (1%) parents/carers completed the adult-directed part of the questionnaire. Evidence of the sample's representativeness is provided by the breakdown of school sectors and socio-economic status (SES) levels, which are in line with Australian population parameters (ABS, 2018, 2021).

Data are shared with another study (Collie et al., 2024) and were collected via an online questionnaire in August 2022. The study period took place amid the COVID-19 pandemic, although all participants attended school in person. Recruitment was carried out using Qualtrics and their associated market research partners, who have access to a broad cross-section of the Australian population. This approach to online recruitment allowed for sampling across the entire country, and the collection of data from both parents (or carers) and their adolescent children. Prospective parent respondents received invitations to participate in the study either via email or app notification.

Parents gave their informed consent and were requested to verify that they had a 13- to 16-year-old child who attended school in person. They were then asked to respond to questions pertaining to their adolescent child (if they had more than one child of this age, they were instructed to select one child). Afterwards, parents were asked to hand over their mobile device to the same adolescent child, who subsequently granted their consent and answered a series of questions directed to them. Respondents who provided swift responses (<1/3 of the median time), provided identical answers to several different items in succession (80% of the survey), answered in a similar manner to oppositely worded item pairs,[1] or shared IP addresses (matched with replicate background characteristics) were excluded from the final sample (Dewitt et al., 2019). The response rate for the study was 80%. Ethics approval was secured from the Human Ethics Research Committee at the author's institution (approval number HC200521).

Measures

Unless specified differently, items used a seven-point scale ranging from 1 (Strongly disagree) to 7 (Strongly agree). Parents reported three of the four outcomes (homework practices, expressive skill, perspective-taking skill), and some background characteristics. All other variables were student reported. Evidence of validity for the scale scores has been established in previous research (Collie, 2022c; Skinner et al., 2008; Soto et al., 2022), and this study offers additional support through the analyses (including measurement invariance testing, see details below). The reliability of the scale scores was assessed by calculating McDonald's omega coefficient and was found to be satisfactory for all factors, as reported in Results.

Need satisfaction

The Social-Emotional Need Satisfaction scale (Collie, 2022d) was used to assess *perceived social-emotional autonomy* (4 items; 'I have choice and freedom in how I behave at school'), *perceived relatedness with students* (4 items; 'My friends at school care about me'), and *perceived relatedness with teachers* (4 items; 'I feel close and connected with my teachers'). Perceived social competence and perceived emotional competence were assessed with the five-item Perceived Social-Emotional Competence Scale – Short (Collie, 2022b). *Perceived social competence* was assessed with three items that tap into perceived competence for assertiveness ('When I need something, I feel confident to ask for help from adults in my life'), tolerance ('I feel capable to be open-minded towards people who have very different views to me'), and social regulation ('I feel capable to act in ways that are considerate towards others'). *Perceived emotional competence* was assessed with two items that tap into emotion regulation ('I feel capable to adjust how I'm feeling if I want to feel less bad about something') and emotional awareness ('I feel confident I can name all the emotions I feel when I'm upset [e.g., disappointed, guilty]').

Hypothesised outcomes

Behavioural disaffection was assessed with four items from Skinner et al. (2008): 'When I'm in class, I just act like I'm working', 'I don't try very hard at school', 'In class, I do just enough to get by', and 'When I'm in class, my mind wanders'. *Homework practices* were assessed with four items adapted from Collie et al. (2019): 'How often does your child do the homework given to them?' 'How often does your child complete the homework given to them?' 'How often does your child spend satisfactory time on the homework given to them?' and 'How often does your child do the homework given to them to the standard expected by their teacher?' The original items were student reported and thus were adapted in the present study to be parent reported. This involved changing the word 'you' in original item to 'your child'. Parents responded on a scale ranging from 1 (Never) to 5 (Always).

Expressive skill and *perspective-taking skill* were assessed with eight items from Soto et al.'s (2022) Behavioural, Emotional, and Social Skills Inventory (BESSI). Items for expressive skill were: How well can your child… 'Explain what they are thinking and feeling', 'Express their thoughts and feelings', 'Tell people how they are feeling', and 'Explain what's on their mind'. Items for perspective-taking skill were: How well can your child… 'Sympathize with other people's feelings', 'Feel compassion for other people', 'Take another person's perspective', and 'Understand how other people feel'. Items were scored from 1 (Not at all) to 5 (Extremely well).

Background characteristics

Students' gender was coded as 0 for male and 1 for female (students who reported their gender as non-binary or other were coded as missing on the gender covariate in the main analyses). Age was recorded as a continuous variable and expressed in years. Language background was coded as 0 for English speaking background and 1 for non-English speaking background. ADHD status was based on parent reports and coded as 0 for no diagnosis and 1 for ADHD diagnosis. The index of relative socio-economic advantage and disadvantage (ABS, 2018), based on parent reports of home postcode, was used to code participants' SES. Higher scores on this index indicate higher SES.

Data analysis

M*plus* 8.9 (Muthén & Muthén, 2021) was used for all analyses, which entailed confirmatory factor analysis (CFA) and structural equation modelling (SEM). Robust maximum likelihood (MLR) was used as the estimator, and full information maximum likelihood was employed to handle missing data ($<$1%). Model fit was judged with the root-mean-square error of approximation (RMSEA) and the comparative fit index (CFI). RMSEA values $\leq$ .08 and CFI values $\geq$ .90 reflect adequate fit (Hu & Bentler, 1999).

Means and standard deviations were calculated for each substantive factor. Tests of measurement invariance were run to ascertain the extent to which the parameters were similar across major participant subgroups by gender (male vs. female), age (13–14 year olds vs. 15–16 year olds), and SES (median split). In order to assess measurement invariance, three models were estimated for each background characteristic:

the configural model, which allowed all parameters to be freely estimated across subgroups; the metric model, which constrained factor loadings across subgroups; and the scalar model, which constrained both factor loadings and item intercepts across subgroups. To determine invariance, increases in RMSEA of $\leq$ .015 and declines in CFI of $\leq$ .01 were used as criteria when comparing two models in the sequence (Chen, 2007; Cheung & Rensvold, 2002). Invariance was concluded when changes in RMSEA and CFI were within those cut-offs.

CFA was next run involving all substantive factors and background characteristics. The substantive factors were entered as latent variables, whereas the background characteristics were entered with loading @1 and residual @0. Correlations were obtained from the CFA and McDonald's omega coefficients were calculated for each substantive factor. Finally, SEM was run using the same specification as the CFA, but with the addition of controls for shared variance and adjusting for background characteristics. Here, the need satisfaction factors were entered as predictors of the four hypothesised outcomes.

Results

Means and standard deviations are shown in Table 1. Table 2 presents the fit indices for the measurement invariance tests, which demonstrate that invariance was achieved across the three background characteristics (ΔRMSEA $\leq$ .015; ΔCFI $\leq$ 0.01). The CFA yielded adequate fit: χ^2 (579) = 1041.99, $p < .001$, RMSEA = 0.040, CFI = 0.95. As shown in Table 1, the reliability coefficients were adequate for all factors. Factor loadings (mean and range) are shown in Table 1, and correlations among substantive factors are shown in Table 3. For correlations among substantive factors and background characteristics, see Supplementary Materials. Focusing on the substantive factors, correlations were as expected. The need satisfaction factors were positively intercorrelated. Behavioural disaffection was negatively associated with all need satisfaction factors and the remaining three outcomes. The three parent-reported outcomes were positively associated with the need satisfaction factors and each other.

The SEM also yielded adequate fit: χ^2 (579) = 1041.99, $p < .001$, RMSEA = 0.040, CFI = 0.95. Table 4 displays beta estimates and *R*-squared estimates for the SEM. Results are also displayed in Figure 2. Starting with substantive factors, perceived

Table 1. Reliability coefficients, descriptive statistics, and factor loadings.

	ω	*M*	*SD*	Factor loadings *M* (range)
Need satisfaction				
Perceived autonomy	0.78	4.61	1.12	0.69 (0.58–0.75)
Perceived social competence	0.76	5.52	0.90	0.72 (0.71–0.73)
Perceived emotional competence	0.73	5.05	1.10	0.76 (0.74–0.78)
Relatedness with students	0.91	5.49	1.09	0.85 (0.81–0.89)
Relatedness with teachers	0.91	5.11	1.14	0.85 (0.82–0.88)
Hypothesised outcomes				
Behavioural disaffection	0.76	3.86	1.22	0.66 (0.52–0.79)
Homework practices (p)	0.94	3.72	1.03	0.89 (0.87–0.91)
Expressive skill (p)	0.92	3.22	0.85	0.85 (0.83–0.87)
Perspective-taking skill (p)	0.89	3.46	0.80	0.82 (0.77–0.85)

Notes: Parent-reported factors are represented by (p). Parent-reported scales were scored from 1 (not at all) to 5 (extremely well). All other scales were scored from 1 (strongly disagree) to 7 (strongly agree).

Table 2. Results from measurement invariance tests.

	RMSEA	CFI
Gender		
Configural	0.046	0.95
Metric	0.046	0.95
Scalar	0.046	0.95
Age		
Configural	0.047	0.95
Metric	0.046	0.95
Scalar	0.046	0.95
SES		
Configural	0.048	0.94
Metric	0.047	0.94
Scalar	0.047	0.94

Note. SES: socio-economic status.

Table 3. Latent correlations among substantive factors in study.

	1.	2.	3.	4.	5.	6.	7.	8.
Need satisfaction								
1. Perceived social-emotional autonomy								
2. Perceived social competence	.52							
3. Perceived emotional competence	.68	.73						
4. Perceived relatedness with students	.56	.70	.59					
5. Perceived relatedness with teachers	.65	.65	.65	.54				
Hypothesised outcomes								
6. Behavioural disaffection	−.30	−.57	−.41	−.41	−.58			
7. Homework practices	.37	.47	.52	.39	.56	−.65		
8. Expressive skill	.51	.53	.64	.47	.46	−.41	.46	
9. Perspective-taking skill	.43	.55	.56	.45	.39	−.33	.43	.68

Note. All correlations significant at $p < .001$.

social competence was associated with lower behavioural disaffection. Perceived emotional competence was associated with more positive homework practices, greater expressive skill, and greater perspective-taking skill. Perceived relatedness with teachers was linked with lower behavioural disaffection and more positive homework practices. Perceived social-emotional autonomy and perceived relatedness with students were not consistently associated with any outcomes. Turning to the covariates, female students were reported by parents as exhibiting more positive homework practices. Older students reported greater behavioural disaffection. Students with a non-English speaking background reported greater social-emotional autonomy. Students with ADHD reported lower levels of all need satisfaction factors, and were reported by parents as exhibiting less positive homework practices and lower expressive skill. Finally, students from higher SES backgrounds were reported by parents as exhibiting more positive homework practice and greater expressive skill.

Discussion

This study investigated the role of social-emotional need satisfaction in relation to academic engagement and social-emotional skills. Results demonstrated that perceived social competence was associated with lower behavioural disaffection, and perceived emotional competence was associated with more positive homework practices, greater expressive skill, and greater perspective-taking skill. Perceived relatedness with

Table 4. Standardised beta estimates.

	Need satisfaction					Hypothesised outcomes			
	Perceived social-emotional autonomy	Perceived social competence	Perceived emotional competence	Perceived relatedness with students	Perceived relatedness with teachers	Behavioural disaffection	Homework practices	Expressive skill	Perspective-taking skill
Covariates									
Gender	0.02	0.06	−0.08	−0.07	0.01	−0.04	0.16***	0.04	0.04
Age	0.08	0.02	0.02	0.04	0.01	0.09*	0.01	0.04	0.07
Language background	0.09*	0.02	0.04	0.05	0.05	−0.01	0.02	−0.03	0.01
ADHD	−0.22***	−0.29***	−0.27***	−0.21***	−0.26***	0.03	−0.17***	−0.11*	−0.10
SES	−0.01	−0.03	0.05	0.01	0.07	−0.02	0.16***	0.11**	0.03
Need satisfaction									
Perceived social-emotional autonomy						0.16	−0.17	0.12	0.08
Perceived social competence						−0.37**	−0.06	0.07	0.28
Perceived emotional competence						0.08	0.34*	0.44***	0.30*
Perceived relatedness with students						−0.03	0.08	0.09	.07
Perceived relatedness with teachers						−0.46***	0.38***	−0.04	−.11
R^2	6.5%	9.5%	8.0%	5.0%	7.8%	43.3%	45.2%	45.2%	38.4%

Note. ADHD: attention-deficit/hyperactivity disorder. SES: socio-economic status. * $p < .05$, ** $p < .01$, *** $p < .001$.

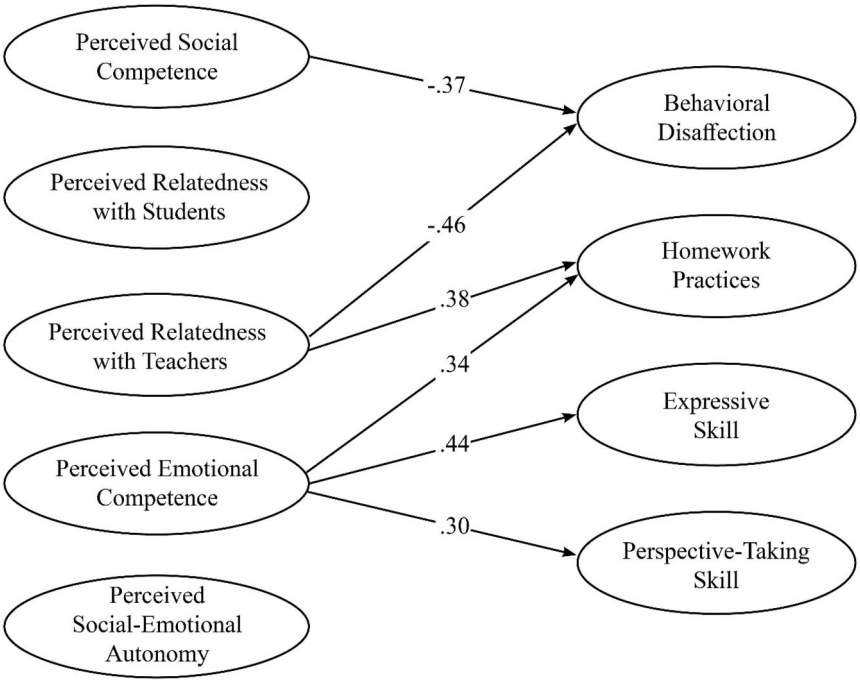

Figure 2. Results from structural equation model. Standardised betas are shown. Only significant paths shown. For all paths (including non-significant paths and those involving covariates), see Table 3.

teachers was associated with lower behavioural disaffection and more positive homework practices. The background characteristics were also associated with the factors in various ways. The results of this study are consistent with the fundamental principles of self-determination theory (Ryan & Deci, 2017) and with SEC conceptualising (Collie, 2020). The findings also advance knowledge in several important ways. This is the first study to examine perceived social-emotional autonomy and competence in relation to academic engagement and social-emotional skills. The findings also yield new knowledge about the unique role of each need satisfaction factor in relation to the hypothesised outcomes.

Notable findings

Students with greater perceived social competence reported lower behavioural disaffection. This finding extends prior research examining social-emotional outcomes (e.g. Kristensen et al., 2021) to show that perceived social competence is also relevant for students' academic (dis)engagement – students who are more confident in their social skills appear less disengaged from their learning. This finding is notable as it underscores the social nature of learning, and the importance of considering social factors when endeavouring to understand academic engagement (Wentzel & Muenks, 2016). Perceived social competence allows students to feel capable in their social interactions, which enables them to mobilise adaptive thoughts and actions aligned with

those perceptions (Ryan & Moller, 2017). This likely results in lower levels of disinterest and apathy relating to learning because students are involved and engaged due to those adaptive thoughts and actions.

Perceived emotional competence was the most consistent predictor in the model. This factor was positively associated with all parent-reported outcomes: homework practices, expressive skill, and perspective-taking skill. Prior research has linked perceived emotional competence with social-emotional behaviours and well-being (Caprara et al., 2008; Parise et al., 2019). The present study extends that understanding to link perceived emotional competence with a broader range of hypothesised outcomes. This finding is noteworthy given much prior research has focused on social predictors (e.g. perceived relatedness; Collie, 2022c), rather than emotional factors. Perhaps perceived emotional competence is linked with more positive homework practices because homework can incite a range of negative emotions – such as frustration from having to do more work after school, or if tasks are too hard with no support after hours (Corno & Xu, 2004). Feeling able to manage one's emotions, then, may allow students to undertake and complete their homework more effectively. Alternatively, it may be that when students feel more emotionally competent, others (e.g. parents) perceive them as being more engaged in their schoolwork (cf. Olivier et al., 2019). Although, the hypothesised model examined here focused on the process established in theory (i.e. Collie, 2020; Ryan & Deci, 2017), it is also possible that there are reciprocal associations. For example, by regularly overcoming the challenges of homework (including any negative emotions), students may learn to deal with those challenges and thus feel more emotionally competent as a result. Future research that tests this association experimentally is needed to ascertain whether there are reciprocal effects.

Turning to the other outcomes, students who report greater perceived emotional competence are more confident in their abilities to regulate and recognise their owns emotions (Collie, 2022a). This confidence helps the students to activate adaptive thoughts and actions (Ryan & Moller, 2017) related to communicating their own thoughts and feelings (expressive skill), as well as in understanding others' thoughts and feelings (perspective-taking skill). As noted above, reciprocal associations should also be tested in future (e.g. because students who self-regulate their emotions more often may come to feel more competent about emotion regulation). Together, the findings involving perceived emotional competence show that this factor is relevant across academic and social-emotional outcomes.

Perceived relatedness with teachers was associated with lower behavioural disaffection and more positive homework practices. These findings align with prior research showing the importance of positive teacher-student relationships for similar outcomes (e.g. Burns et al., 2019; Martin et al., 2009; Skinner et al., 2008). The results additionally highlight the unique role of relatedness with teachers beyond the role of the other social-emotional need satisfaction factors. In particular, it is noteworthy that the association to homework practices occurred even with the use of different respondents (students reported on relatedness, parents reported on homework). Together, the findings underscore the importance of perceived relatedness with teachers for positive academic outcomes within and outside the classroom (Gregory & Korth, 2016).

Perceived social-emotional autonomy and perceived relatedness with students were associated with all hypothesised outcomes in the bivariate correlations (CFA), but no associations remained significant after controls for shared variance and background characteristics were included in the SEM. Prior research has shown that perceived social-emotional autonomy is associated with lower negative affect (Collie, 2022c), suggesting this factor may be more relevant for well-being outcomes. The result involving perceived relatedness with students aligns with prior research, which demonstrated that beyond the role of perceived social competence, there was a non-significant association from this factor to behavioural and well-being outcomes (Collie, 2022c). Both perceived relatedness with students and perceived social competence relate to interpersonal interactions. The finding, then, suggests that it is perceived social competence that explains unique variance in relation to the outcomes. Future research is important to determine whether the two need satisfaction variables play unique roles in relation to other outcomes.

For the background characteristics, ADHD diagnosis was the variable that played the most prevailing role. Students with ADHD reported lower levels of all need satisfaction factors, and were reported by parents as exhibiting significantly less positive homework practices and lower expressive skill than students without ADHD. The results may be explained by the fact that students with ADHD often report less favourable learning contexts (Martin et al., 2017), and given several of the factors are relevant to executive functioning (i.e. homework practices, expressive skill), which is impaired in students with ADHD (Barkley, 2018). The results underscore the importance of supporting students with ADHD in relation to these factors (e.g. Hare et al., 2021).

Implications for practice

Findings suggest that teachers may want to support students' perceived social and emotional competence, as well as their perceived relatedness with teachers. Starting with perceived social competence, teachers can ask students to think about a recent situation for which they could have regulated their actions differently, generate ideas for how they could respond more effectively in future, put those ideas in practice, and, then evaluate and refine their ideas for next time (e.g. Boekaerts & Pekrun, 2016). Teaching practices to support students' perceived emotional competence include encouraging students: to take actions to prevent situations that create undesired emotions (e.g. avoiding study procrastination to reduce anxiety about an upcoming test), to reframe the situation if it is unavoidable (e.g. reminding oneself that the test is not high-stakes; Gross, 2015), and to reflect on the emotional accounts of people they are learning or reading about (Brewer & Phillippe, 2022). For boosting perceived relatedness, teachers can demonstrate care and concern for students and their development, dedicate time and efforts to students, and deal with classroom issues democratically (Skinner & Belmont, 1993).

Limitations and future directions

The first limitation of the present study is that the data were cross-sectional in nature, meaning that causal ordering could not be determined. The construct positioning in

the hypothesised model was determined from theory, and is supported by experimental and longitudinal research in academic domains (Cheon et al., 2018; Jang et al., 2016). However, reciprocal associations are also possible (Wray-Lake et al., 2019), and studies with longitudinal and experimental designs are now needed to test for the presence of reciprocal effects within social-emotional domains. Second, three of the outcomes were parent reported. This is a strength given that it reduces concerns about common-method bias, and also because it helped to ensure that students' own reports of social-emotional need satisfaction were disentangled from the outcomes. In Australia, parents typically attend two or more parent-teacher interviews per year where they learn about their child's progress, and are also informed outside of those interviews if their child does not meet the homework expectations. At the same time, some parents may have difficulty in accurately assessing students' homework practices, as well as the social-emotional skills. Thus, it is also important to triangulate these findings with student self-reports and teacher reports to understand the role of social-emotional need satisfaction across these different angles. Third, it is not possible to eliminate the possibility of sampling bias since the parent participants voluntarily enrolled to receive information regarding research studies focusing on adolescents. It is recommended that future research be conducted using alternative samples and recruitment methods to ascertain whether the results of the current study are corroborated. Fourth, the academic outcomes in the present study were related to behavioural (dis)engagement given the desire to focus on behaviourally-relevant outcomes. In future research, it will be important to examine other academic outcomes, such as motivation and achievement. Fifth, ADHD was included as a covariate in the current study because it is the most prevalent neurodevelopmental disorder among students. However, future studies should also consider other social-emotional conditions/disorders (e.g. autism spectrum disorder). Finally, Collie (2022a) points out that dimensions of perceived social-emotional competence may not necessarily be universal and could be accompanied or substituted by other dimensions in different contexts and cultures. As such, it is recommended that future research explores these dimensions in diverse settings and evaluates whether other dimensions should be taken into account. Moreover, the present study assessed broad factors of perceived social competence and perceived emotional competence. In future research, it will be important to assess subdimensions of the two factors to further nuance understanding (e.g. see Collie, 2022a).

Conclusion

The primary objective of this study was to expand understanding of social-emotional need satisfaction by examining it in relation to novel academic and social-emotional factors. The results demonstrated that perceived social competence, emotional competence, and relatedness with teachers were linked with the hypothesised outcomes. Specifically, the factors of need satisfaction exhibited unique associations with the outcomes, providing knowledge relevant for intervention development. Taken as a whole, the study findings provide support for the importance of considering social-emotional need satisfaction in research and practice aimed at fostering positive academic, social, and emotional outcomes among students.

Note

1. Item pairs were from the broader research project and were not examined in this study. The items assessed conflicting motives for engaging in social-emotional behaviors (e.g., "…because I believe this is important to do" and "…actually I don't, because I don't see the point of putting more effort in;" Collie, 2022a). Participants who answered alike to item pairs were excluded (i.e., [strongly] agreeing or disagreeing with both items).

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The ethics approval received for this study does not allow for data sharing.

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